PROBLEM DOMAIN INTEREST – ST-01 (SUSTAINABILITY)

PROBLEM STATEMENT NO : ST-08

-AIR QUALITY MONITORING AND PUBLIC ALERT SYSTEM

- **TEAM MEMBERS :**

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1. Introduction

- Air pollution is a major environmental and health issue affecting cities worldwide.
- The lack of real-time awareness and delayed communication leads to increased health risks.
- This project aims to provide a system that monitors air quality and alerts users instantly.

2. Problem Understanding

- Air pollution in major cities like Delhi, Kanpur, and Patna has become a serious public health issue, especially during winter.
- Although air quality sensors exist, the data is not easily accessible or understandable to the general public in real time.
- This leads to people unknowingly exposing themselves to harmful conditions.
- Schools, hospitals, and authorities also lack timely alerts to take preventive actions.
- Hence, there is a need for a system that delivers real-time air quality data along with clear alerts and health advisories to citizens and officials.

3. Idea Clarity

- The proposed solution is an **Air Quality Monitoring and Public Alert System** that collects data from air quality sensors and APIs, processes it, and displays it through a user-friendly dashboard.
- The system provides real-time AQI updates, alerts for high pollution levels, and health recommendations.

- It includes features like trend analysis, daily reports, complaint submission, and role-based access for public users and government authorities.
- The goal is to make air quality information simple, accessible, and actionable

4. Feasibility

- The solution is highly feasible within a hackathon environment as it uses free and open-source technologies such as HTML, CSS, JavaScript, and APIs like OpenWeather.
- The frontend can be developed quickly, and backend integration can be partial or simulated.
- Real-time data fetching, visualization using Chart.js, and basic alert mechanisms are easy to implement.
- The system can also be scaled later by integrating IoT sensors and cloud storage, making it practical for real-world deployment.

5. Innovation

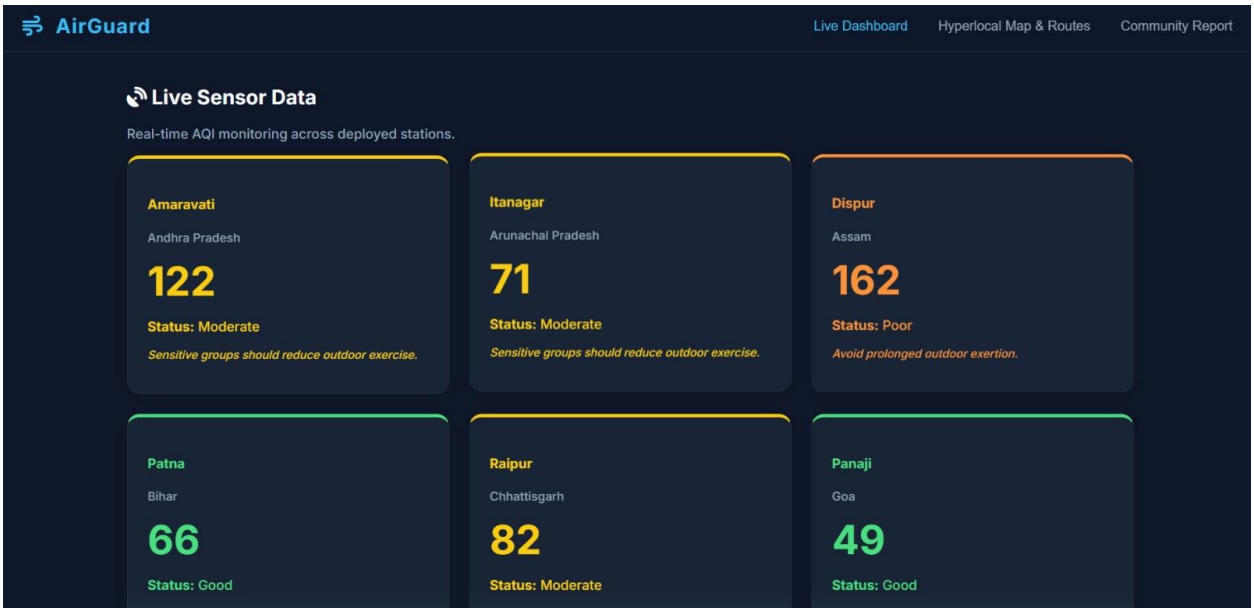
- The innovation lies in combining real-time air quality monitoring with intelligent alert systems and user interaction features.

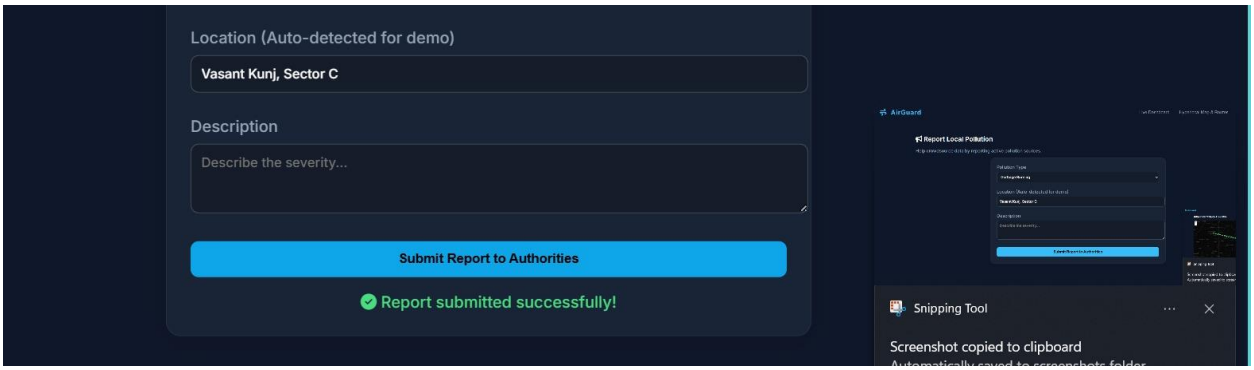
- Unlike existing systems, this solution focuses on **user-friendly visualization**, **instant notifications**, and **health advisory integration**.
- Features like location-based AQI detection, trend prediction, public complaint submission, and admin control panels enhance usability.
- The inclusion of responsive design and smart alerts ensures better engagement and awareness among users.

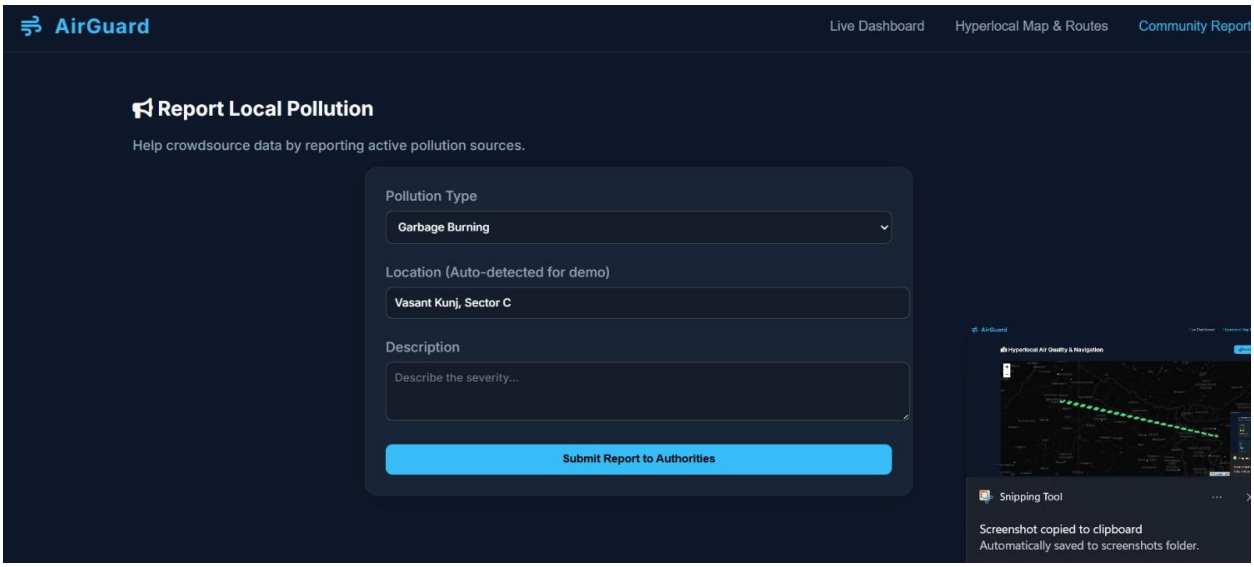
6. Team Planning

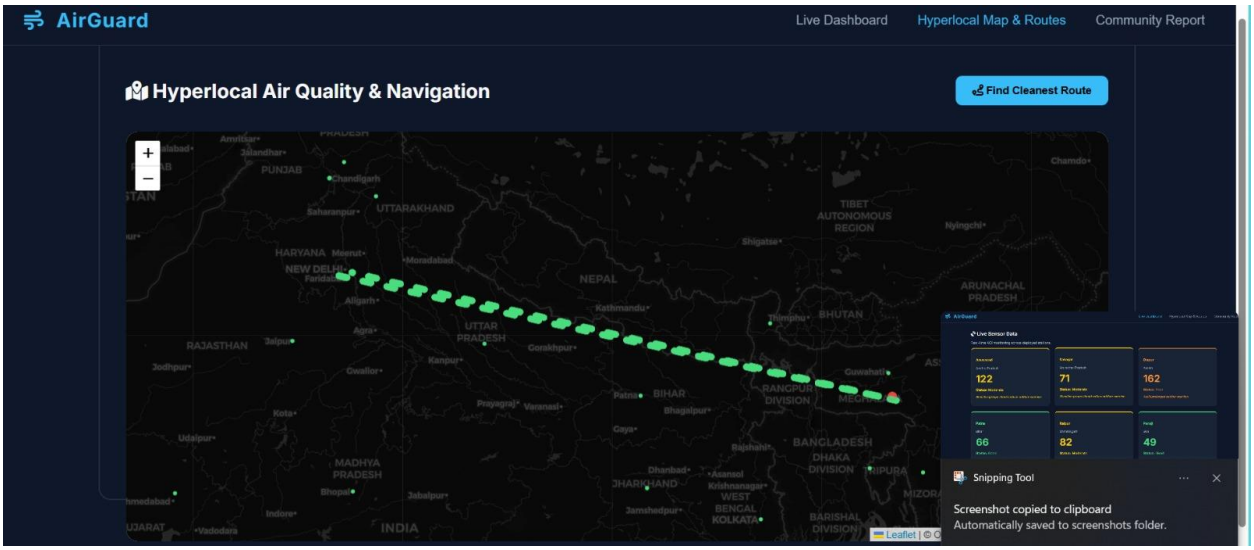
- The project can be effectively executed by dividing tasks among team members.
- One member can handle frontend UI/UX design, another can work on API integration and backend logic, while a third member focuses on data visualization and alerts system.
- Documentation and presentation can be managed collaboratively.
- Regular coordination ensures timely completion within the hackathon timeframe.
- This structured approach improves efficiency, code quality, and overall project delivery.

Output Sreenshots :






AirGuard
Live Dashboard
Hyperlocal Map & Routes
Community Report
Report Local Pollution
Help crowdsource data by reporting active pollution sources.
Pollution Type
Garbage Burning
Location (Auto-detected for demo)
Vasant Kunj, Sector C
Description
Describe the severity...
Submit Report to Authorities
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.


AirGuard
Live Dashboard
Hyperlocal Map & Routes
Community Report
Hyperlocal Air Quality & Navigation
Find Cleanest Route
PUNJAB
UTTARAKHAND
HARYANA
NEW DELHI
NEPAL
TIBET AUTONOMOUS REGION
ARUNACHAL PRADESH
BHUTAN
RAJASTHAN
UTTAR PRADESH
BIHAR
MADHYA PRADESH
BANGLADESH
INDIA
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71
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82
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Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.